# Project Concept Note

MoodBuddy – AI-Powered Mood Check-In Web Application

## 1. Project Title and Application Name

**Project Title:** MoodBuddy – A Lightweight AI-Powered Mood Check-In Companion

**Application Name:** MoodBuddy

MoodBuddy is a web-based mood check-in application that allows users to quickly express how they are feeling through fun, low-effort interactions. An integrated AI model then generates a short, warm, and encouraging response, along with simple playful activities designed to gently lift the user's mood.

## 2. Problem Statement / Objective

In daily life, people often experience shifting emotions but rarely pause to acknowledge or reflect on how they feel. Most existing mood-tracking or wellness applications are either overly clinical, time-consuming to use, or framed around medical/diagnostic language that can feel intimidating or inappropriate for casual, everyday emotional check-ins.

The objective of MoodBuddy is to provide a simple, fast, and non-clinical way for users to check in with their emotions in under a minute, receive a warm and encouraging AI-generated response, and be gently nudged toward a small, playful activity that helps improve their mood. The application deliberately avoids any diagnostic or medical terminology (such as "depression" or "anxiety disorder") and instead focuses purely on validation, encouragement, and light positive action.

## 3. Target User and Use Case

**Target Users:** Students, working professionals, and general users — particularly those seeking a quick, casual, and low-commitment way to reflect on their day-to-day emotional state without engaging with a full-fledged mental health platform.

**Primary Use Case:** A user opens MoodBuddy once or a few times a day (for example, at the start of the day, after a stressful event, or before bed). Within a few taps, they select how they feel using a mood option, a color, and a set of descriptive tags, optionally add a short text note, and receive an AI-generated, empathetic summary along with a suggested mood-boosting activity. Over time, the user can view their mood history on a simple dashboard to notice patterns and trends.

## 4. LLM Model and API Used

MoodBuddy uses the **Google Gemini API** (free tier), specifically a lightweight model such as **gemini-1.5-flash** or **gemini-2.0-flash**, to generate the mood response text.

The AI is called exclusively from the backend server, never from the frontend, so that the API key is never exposed to end users or committed to version control. Responses are streamed from Gemini's streaming endpoint to the backend and then relayed to the frontend using Server-Sent Events or chunked responses, allowing the AI's reply to render progressively on the Mood Result page rather than appearing all at once.

## 5. Key Features of the Application

- **Multi-Step Mood Check-In Flow:** A quick, four-part flow — multiple-choice mood selection, a color palette mood picker, multi-select descriptive tags, and an optional short text note (capped at 200 characters).
- **AI-Generated Mood Response:** A warm, non-clinical, streaming AI response that includes a short mood summary, a line of encouragement, a suggested activity, and a brief affirmation.
- **Four Mood-Boosting Activities:** Breathing Rainbow (guided breathing animation), Music Suggestions (curated calming audio categories), Mood Doodle (a simple drawing canvas), and Color Match Rush (a quick reaction mini-game).
- **Mood History and Dashboard:** A minimal, single-table mood history with a simple last-7-days / last-30-days chart view so users can see trends over time.
- **Profile Page:** Displays the user's name, avatar, join date, streak, and total check-ins.
- **Secure, Decoupled Architecture:** A clearly separated frontend and backend, with the LLM API key and all secrets held only on the backend and never exposed client-side.
- **Responsive, Calming Design:** A soft, premium, glassmorphism-inspired interface with gentle animations, rounded cards, and large tappable buttons, built to feel calm and friendly rather than clinical.

## 6. Expected User Experience and Outcomes

MoodBuddy is designed to feel like a quick, friendly check-in rather than a chore or a clinical assessment. The end-to-end experience — from opening the app to receiving an AI response — is expected to take under a minute, with a calm, visually pleasing interface that encourages regular, low-pressure use.

Expected outcomes for users include:

- A quick sense of being heard and validated through a warm, personalized AI response.

- A gentle, actionable nudge toward a small positive activity suited to their current mood.
- Increased self-awareness over time through a simple visual history of mood trends.
- A consistently non-clinical, non-judgmental experience that feels supportive rather than medical or diagnostic.
- A lightweight habit of regular emotional check-ins that is easy to sustain because each interaction is short, playful, and low-effort.

In summary, MoodBuddy aims to be a small, delightful, everyday companion for emotional check-ins — combining a fast, playful interaction flow with a secure, well-structured technical foundation and a strictly non-clinical, supportive tone throughout.